CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 04/12/2018 **Day: Tuesday** **Time: 10:00am to 10:30am** **Total Marks: 20**

Student's ID No.
Student's Name: ___________________________
Student's Sign : ___________________________
Supervisor's Name : ________________________
Supervisor's Sign: __________________________

Answer Sheet for Multiple Choice Questions (MCQs),True or false and Match the following

Examination : BOP- December 2018

Semester: III

Section : I

Course Code : BOP303

Course Name: Optometric Optics I

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself &**SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs) **(10 × 1 = 10)**
(Encircle the correct answer)

1. **Pin cushion distortion is seen in ___**

   a. Minus lens
   b. Plus lens
   c. Glass slab
   d. Prism

2. **Longest wavelength is the visible spectrum is of _____ color**

   a. Red
   b. Violet
   c. Yellow
   d. Blue

3. **Spherical equivalent for the prescription of -0.50/-2.00×10 is __**
   - **a.** -1.00Ds
   - **b.** -2.00Ds
   - **c.** -1.50Ds
   - **d.** -3.00Ds
4. **Spherical equivalent for prescription of +2.00/+5.75×90 is__**
   - **a.** +4.75Ds
   - **b.** +4.87Ds
   - **c.** +3.87Ds
   - **d.** +3.75Ds
5. **Which of the following is correct?**
   - **a.** Distance power + Near power = Near Rx
   - **b.** Near power – Distance power = Near addition
   - **c.** Distance power – Near power = Near addition
   - **d.** Near power – Distance power = Near Rx
6. **Shape of cornea is____**
   - **a.** Plano convex
   - **b.** Bi convex
   - **c.** Bi concave
   - **d.** None of the above
7. **Power of lens with focal length 50cm is__**
   - **a.** 2.50D
   - **b.** 2.00D
   - **c.** 3.00D
   - **d.** 3.50D
8. **A lens clock is calibrated for glass of index _____**
   - **a.** 1.525
   - **b.** 1.530
   - **c.** 1.535
   - **d.** 1.515
9. **What does the root word Phot means?**
   - **a.** Light
   - **b.** Color
   - **c.** Image
   - **d.** Disease
10. **Front vertex power is also known as________**

    - **a.** Equivalent power
    - **b.** Neutralizing power
    - **c.** Approximate power
    - **d.** None of the above

Identify the sentence as True or False: **(5 × 1 = 5)**
(Please tick mark the answer)

	True	False
11. Base curve is always weaker/flatter than the cross curve.	☐	☐
12. Higher the Abbe value, the more chromatic aberration present in the lens.	☐	☐
13. 1 degree is equal to 1.75 prism diopters.	☐	☐
14. Fresnel prism is made up of PVC material	☐	☐
15. Plus lens is thinner at centre than in the periphery.	☐	☐

Match the following **(5 × 1 = 5)**

16. Neutralizing power		**A.** Dispersion
17. VIBGYOR		**B**. Front surface power
18. 1.525		**C**. Fresnel prism
19. Measuring phoria & fusional vergence		**D**. Distortion
20. Petzval's surface		**E**. Back surface power
		F. Risley prism
		G. Curvature of Image

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BOP Examination
December 2018
BOP303 Optometric Optics I

Date: 04/12/2018 **Time: 10:30am to 1:00pm** **Maximum Marks: 50**

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself&**SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q.1. Short answers (10 out of 11) **(10 × 2 = 20)**

- **a.** Convert 2 cylinder form into optical cross (1) +4.00×20, -3.00×110, (2) -1.00×100, -11.00×10
- **b.** A certain lens of index 1.523 has a convex spherical surface. The sag of the front surface is 1mm for a chord whose length is 20mm. what is power of front surface?
- **c.** Explain Base curve and Cross curve.
- **d.** A +6.50D lens before the right eye is decentered 3mm nasalward. What amount of prism is induced and what is the base orientation?
- **e.** Write down formula of front and back vertex power.
- **f.** Explain about Resultant horizontal prismatic effects.
- **g.** Describe spherical lens form.
- **h.** Convert spherocylindrical form into optical cross (1)+2.00/+1.00×180 (2)-4.00/-3.00×90
- **i.** What is longitudinal chromatic aberration?
- **j.** Define Equivalent power.
- **k.** Define (1) Prism (2) Refracting angle of prism

Q.2. Short Notes (4 out of 5) **(4 × 5 = 20)**

- **a.** Explain in detail about chromatic aberration.
- **b.** Explain conoid of sturm diagram.
- **c.** Explain in detail about spherical aberration
- **d.** Write a note on Prentice's rule.
- **e.** +1.00/+2.00×180, Base curve=+6.00. Find out the Fraction value using toric transposition steps.

Q.3. Essay **(1 × 10 = 10)**

- **a.** Fresnel Press on prism